

Nacef Guessaymi

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EDUCATION

University of Pittsburgh Aug 2018 — Current
PhD, Molecular Biophysics and Structural Biology Pittsburgh, PA

- Relevant Coursework: Molecular Biophysics, Neural Signal Processing, Computational Chemistry

Temple University Philadelphia, PA
Bachelor of Science, Biophysics Aug 2013 — May 2017

- Relevant Coursework: Thermodynamics and Statistical Mechanics, Mathematical Physics

SKILLS

- **Languages:** Python, MATLAB, Rust, SQL, Bash
- **Scientific Computing:** NumPy, SciPy, Pandas, Polars, OpenMM, RDKit
- **Machine Learning:** scikit-learn, XGBoost, model inference, descriptor-based prediction
- **Infrastructure:** Linux, SLURM, Docker/Singularity, Pixi, Conda, Git
- **Quantitative Methods:** Signal Processing, Statistical Modeling, Stochastic Simulation, Data Structures & Algorithms

WORK EXPERIENCE

PhD Candidate August 2024 — Expected Dec 2026
University of Pittsburgh Pittsburgh, PA

- Developed Python-based scalable computational pipelines for molecular docking, simulation, and trajectory analysis
- Designed and implemented a modular OpenMM-based software package for reproducible biomolecular energy minimization
- Performed large-scale stochastic simulations using high-performance computing environments

Graduate Researcher Jan 2022 — Aug 2024
Carnegie Mellon University Pittsburgh, PA

- Built MATLAB control systems for synchronized optogenetic stimulation and electrophysiology acquisition
- Developed automated signal-processing and image-analysis pipelines for neural probe characterization
- Designed quantitative workflows for analyzing insertion-force mechanics in neural interfaces

Scientist II Jan 2021 — Jan 2022
Eurofins PSS Gaithersburg, MD

- Performed high-throughput quantitative analysis and validation of biologic therapeutic assays under GMP standards
- Processed and analyzed large experimental datasets for accelerated COVID-19 therapeutic programs

Graduate Researcher Jun 2019 — Jan 2021
Carnegie Mellon University Pittsburgh, PA

- Extracted motion features from live-cell microscopy using MATLAB image tracking
- Extracted and analyzed cellular motion features to characterize force-dependent responses under cytoskeletal perturbations

Graduate Researcher Aug 2018 — Jun 2019
University of Pittsburgh Pittsburgh, PA

- Converted gel assay images into quantitative datasets
- Fit binding curves in GraphPad Prism to estimate interaction parameters from biochemical assay data
- Constructed a chimeric protein structure from template models to support molecular dynamics simulation setup

Undergraduate Researcher Apr 2015 — Aug 2018
Temple University Philadelphia, PA

- Automated super-resolution microscopy workflows using ImageJ, Perl, and analysis macros
- Quantified single-molecule motion from live-cell imaging using MATLAB diffusion and MSD analysis
- Reconstructed 3D spatial maps from PSF localization data using Amira